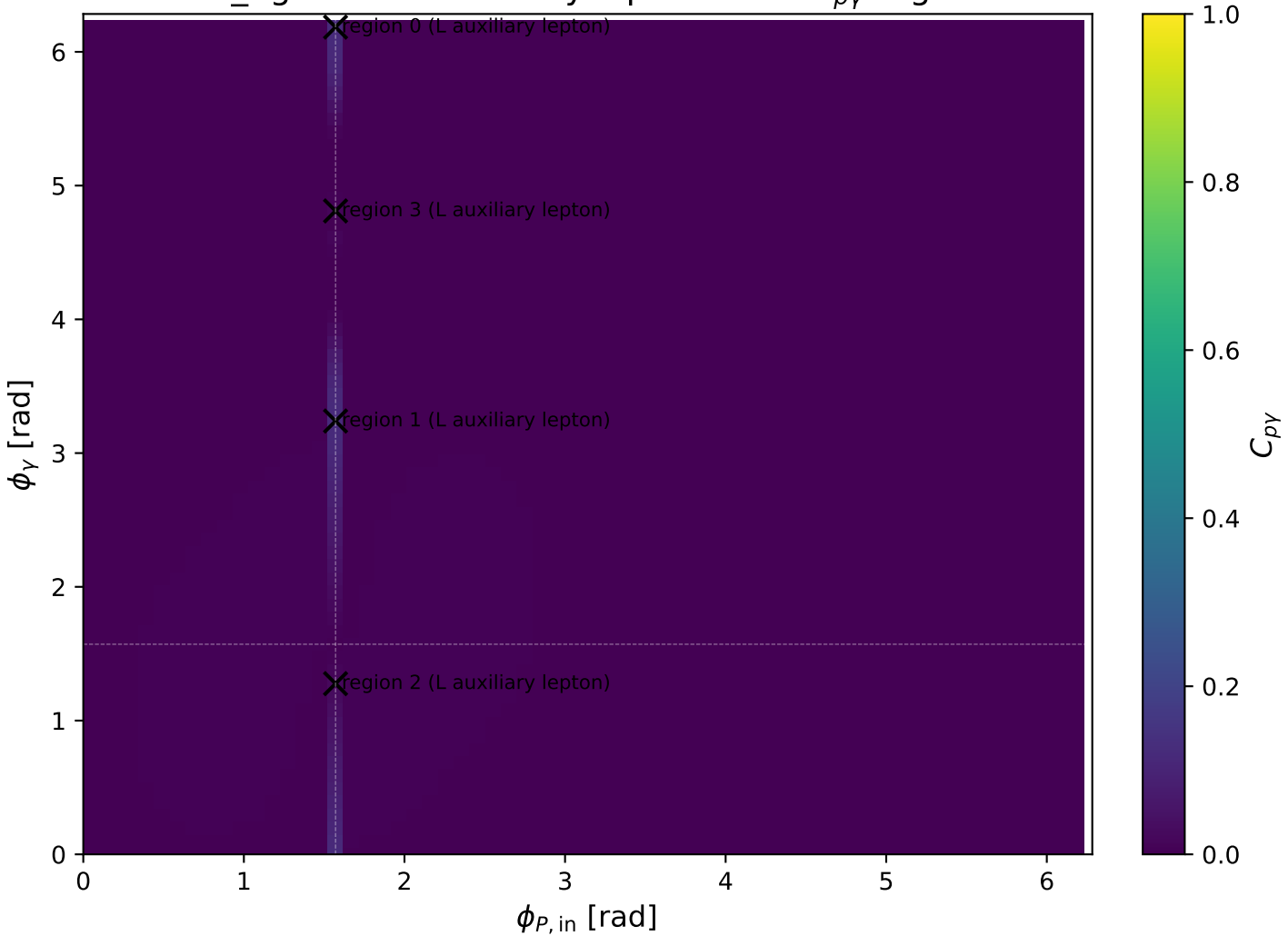
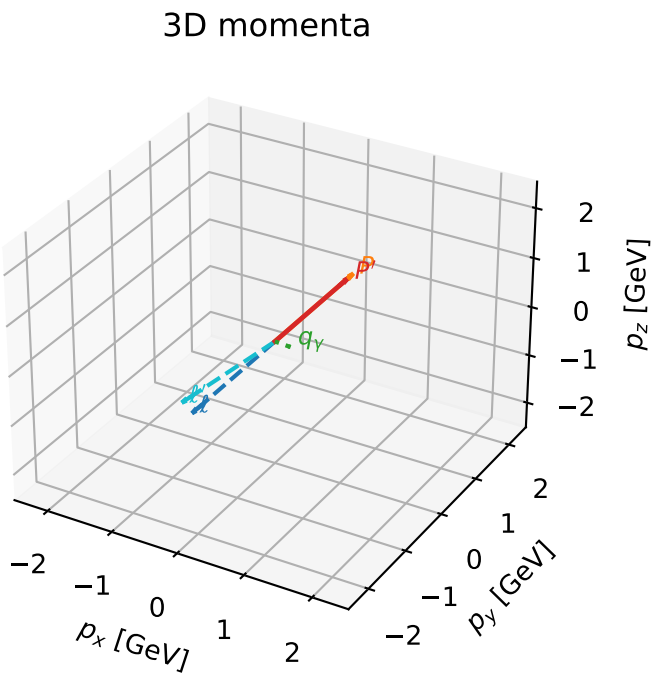


low\_Egamma L auxiliary lepton: max  $C_{p\gamma}$  regions



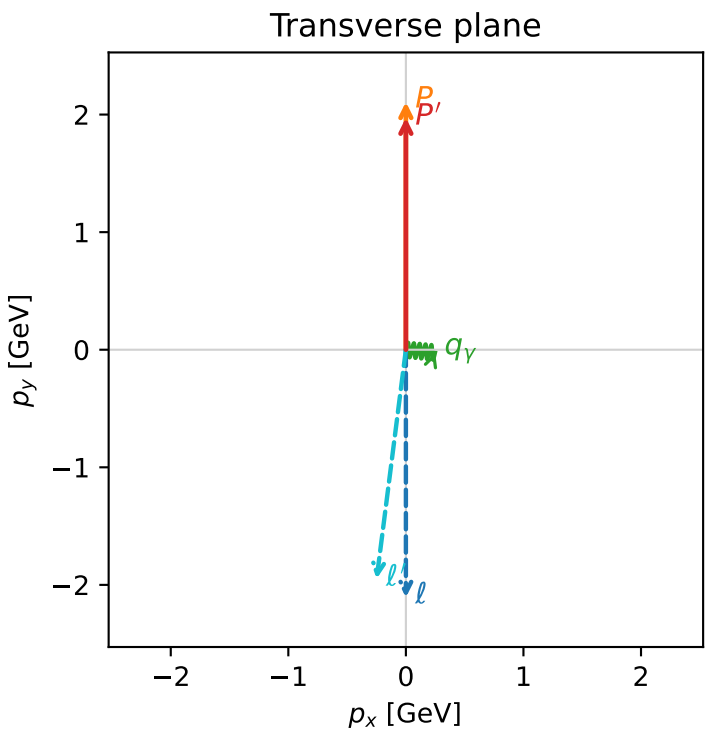
L auxiliary lepton characteristic max  $C_{p\gamma}$  configuration



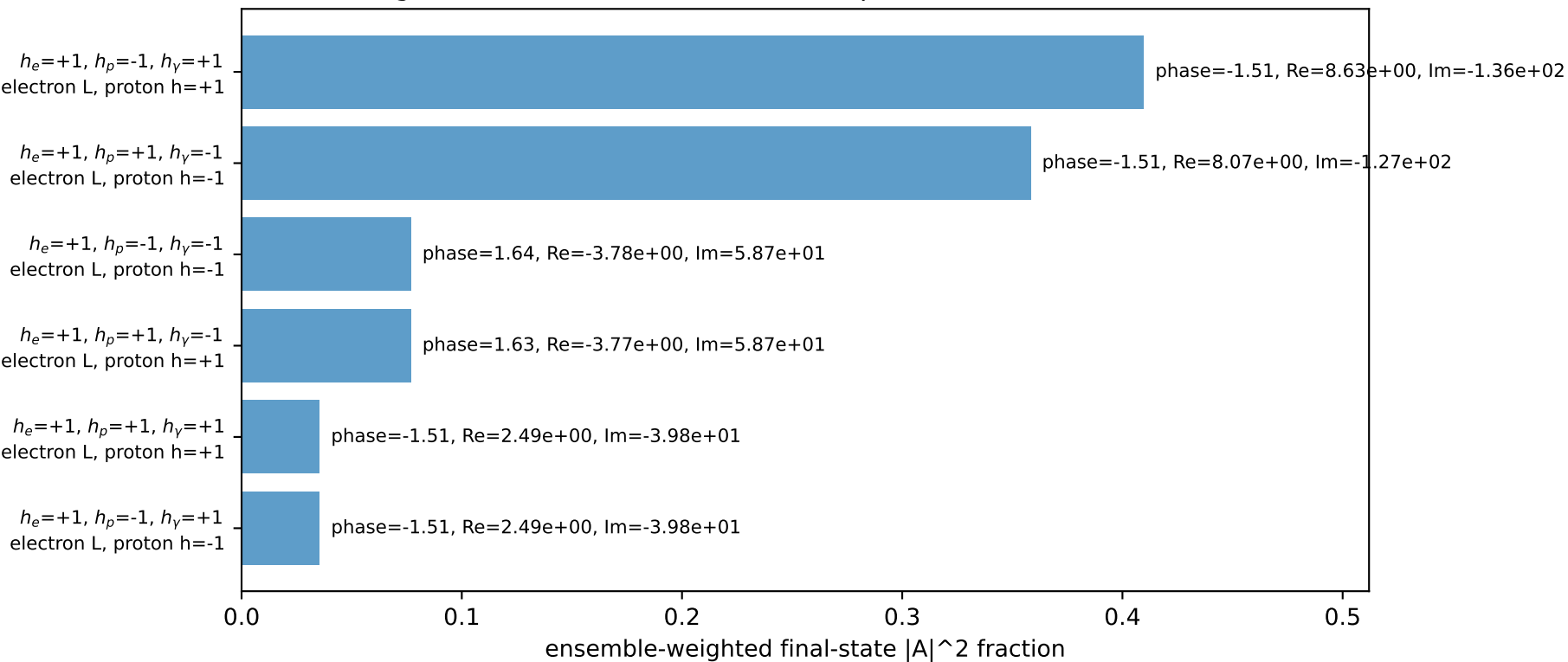
c\_p\_gamma\_low\_egamma\_l\_lepton\_region\_0 C\_{p\gamma}=0.120212  
region: low\_Egamma\_L\_lepton\_region\_0 final-pair delta\_xy=1.66897 rad  
outgoing-state purity: 0.505902  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.97282$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=6.18501$   
 $Q^2=0.0706362$ ,  $x_B=0.0560804$ ,  $t=-0.00314184$ ,  $W^2=2.06876$ ,  $y=0.0641451$   
 $\theta(\ell', \gamma)=91.6522$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.3322$   $p_x= -0.0000$   $p_y= -2.1069$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 0.0000$   $p_y= 2.1069$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2041$   $p_x= -0.2488$   $p_y= -1.9483$   $p_z= 0.0000$   
 $P'$   $E= 2.1845$   $p_x= 0.0000$   $p_y= 1.9728$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2488$   $p_y= -0.0245$   $p_z= 0.0000$

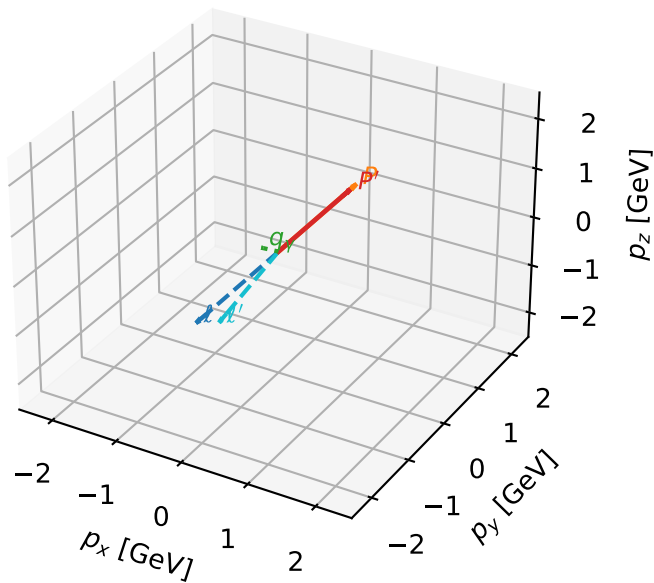


Leading initial-ensemble/final-state components (N=6, retained=99.2%)



# L auxiliary lepton characteristic max $C_{p\gamma}$ configuration

3D momenta

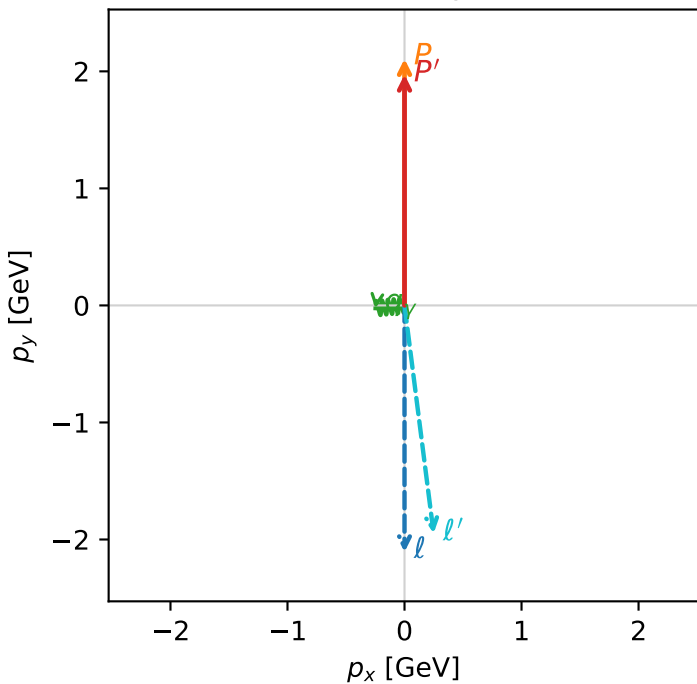


c\_p\_gamma\_low\_egamma\_l\_lepton\_region\_1 C\_{p\gamma}=0.120212  
 region: low\_Egamma\_L\_lepton\_region\_1 final-pair delta\_xy=1.66897 rad  
 outgoing-state purity: 0.505902  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

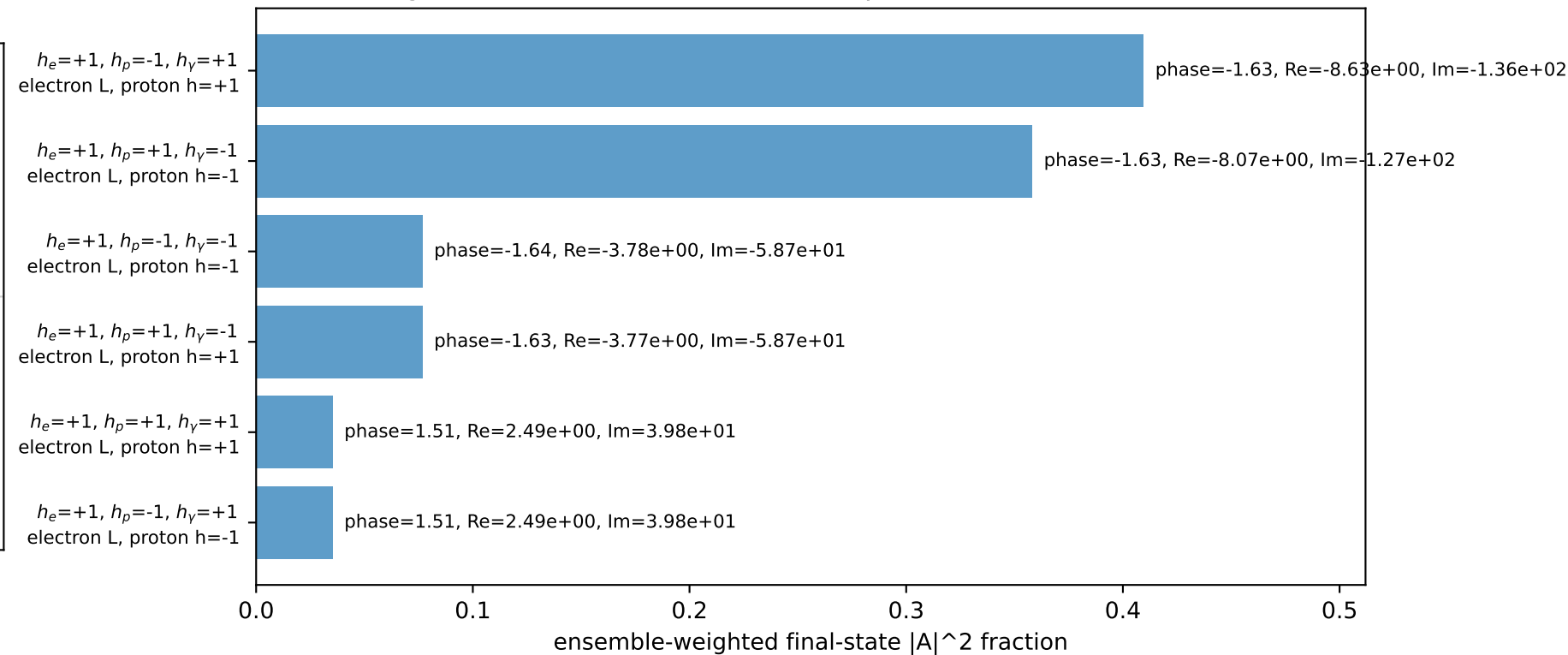
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=1.97282$   
 $\theta_{in}=1.57079$ ,  $\phi_i=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=3.23977$   
 $Q^2=0.0706362$ ,  $x_B=0.0560804$ ,  $t=-0.00314184$ ,  $W^2=2.06876$ ,  $y=0.0641451$   
 $\theta(\ell', \gamma)=91.6522$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.3322$   $p_x= -0.0000$   $p_y= -2.1069$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 0.0000$   $p_y= 2.1069$   $p_z= 0.0000$   
 $\ell'$   $E= 2.2041$   $p_x= 0.2488$   $p_y= -1.9483$   $p_z= 0.0000$   
 $P'$   $E= 2.1845$   $p_x= 0.0000$   $p_y= 1.9728$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2488$   $p_y= -0.0245$   $p_z= 0.0000$

Transverse plane

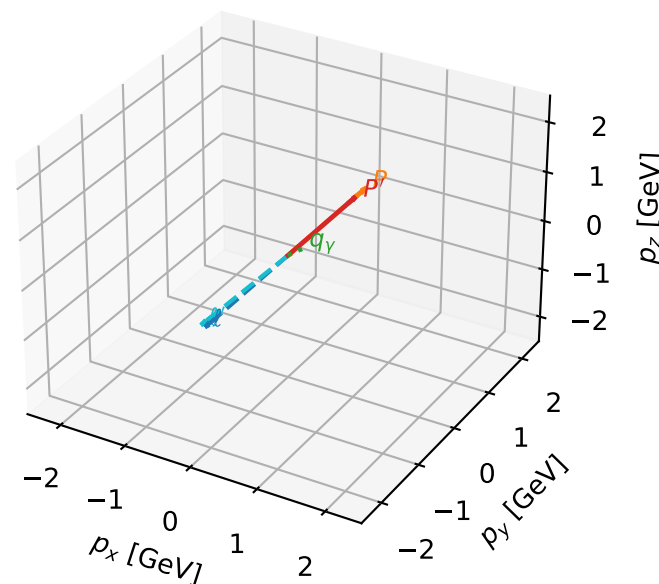


Leading initial-ensemble/final-state components (N=6, retained=99.2%)



# L auxiliary lepton characteristic max $C_{p\gamma}$ configuration

3D momenta

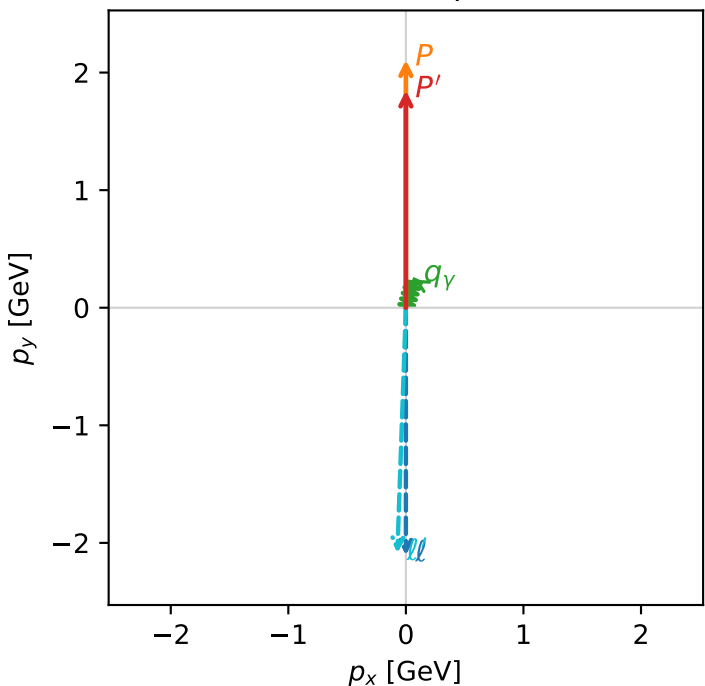


c\_p\_gamma\_low\_egamma\_l\_lepton\_region\_2 C\_{p\gamma}=0.0225655  
region: low\_Egamma\_L\_lepton\_region\_2 final-pair delta\_xy=0.294524 rad  
outgoing-state purity: 0.500277  
kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

s=21.5158,  $\sqrt{s}$ =4.63852,  $|\vec{P}|$ =2.10694,  $|\vec{P}'|$ =1.8483  
 $\theta_{in}$ =1.57079,  $\phi_\ell$ =4.71239,  $\phi_P$ =1.5708  
 $E_\gamma$ =0.25,  $\phi_\gamma$ =1.27627  
 $Q^2$ =0.00537493,  $x_B$ =0.0341558,  $t$ =-0.012321,  $W^2$ =1.03183,  $y$ =0.00801412  
 $\theta(\ell', \gamma)$ =165.116 deg

four-momenta [E, p\_x, p\_y, p\_z] GeV:  
 $\ell$  E= 2.3322 p\_x= -0.0000 p\_y= -2.1069 p\_z= -0.0000  
 $P$  E= 2.3063 p\_x= 0.0000 p\_y= 2.1069 p\_z= 0.0000  
 $\ell'$  E= 2.3158 p\_x= -0.0726 p\_y= -2.0875 p\_z= 0.0000  
 $P'$  E= 2.0727 p\_x= 0.0000 p\_y= 1.8483 p\_z= 0.0000  
 $q_\gamma$  E= 0.2500 p\_x= 0.0726 p\_y= 0.2392 p\_z= 0.0000

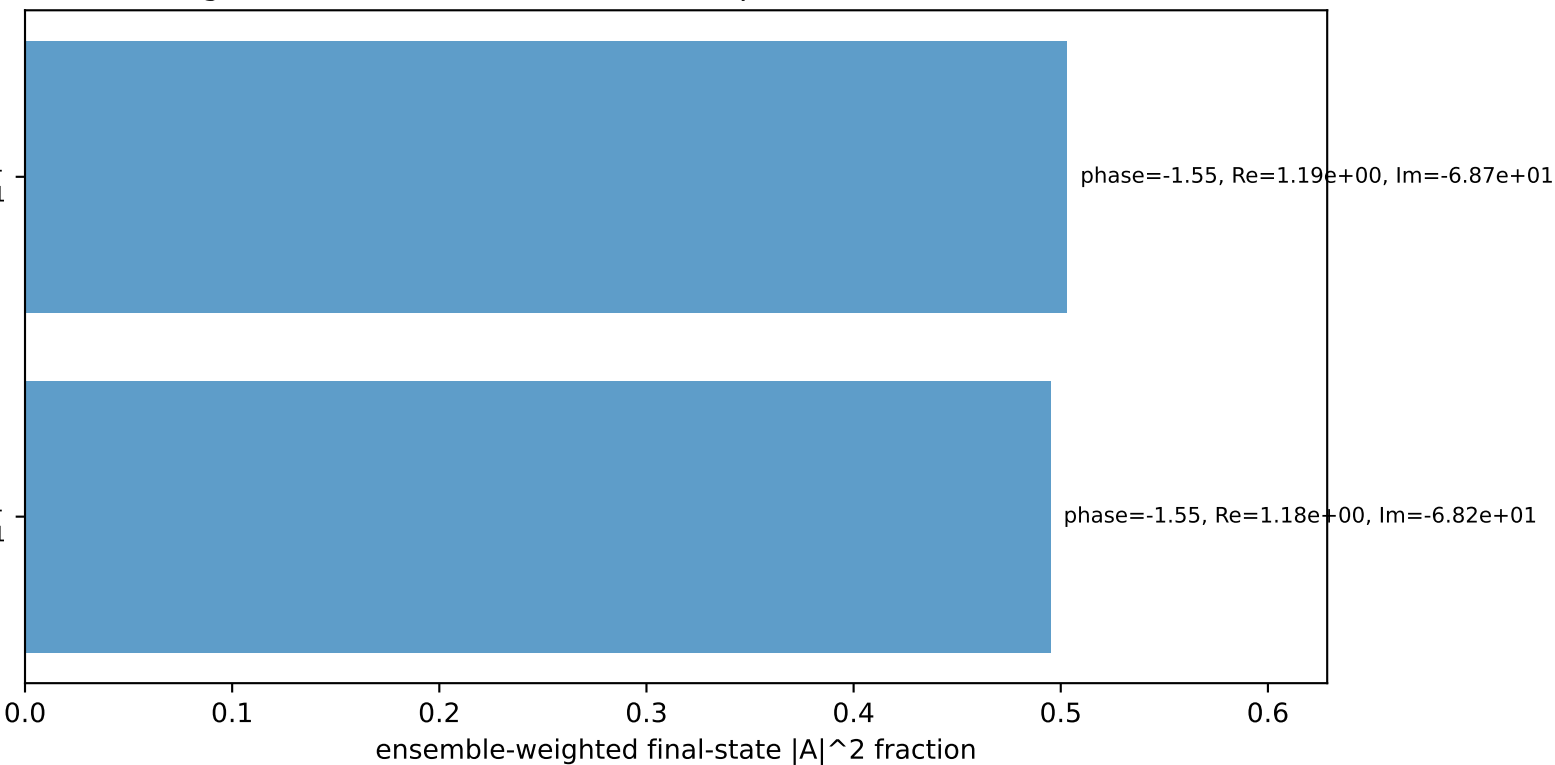
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$   
electron L, proton h=+1

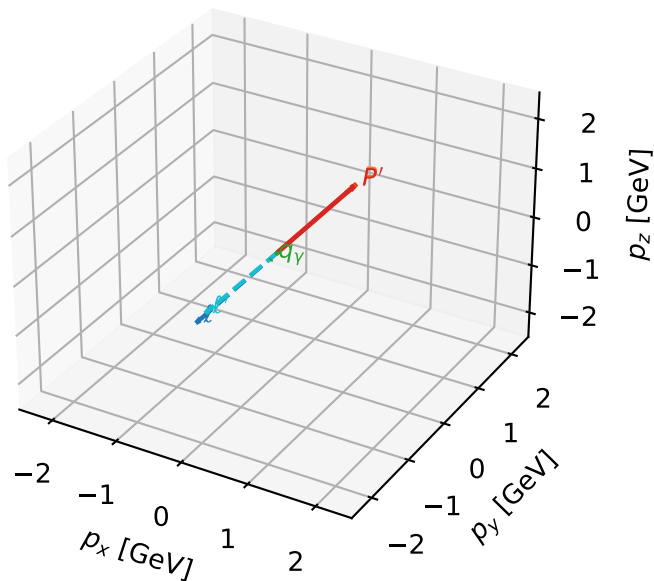
$h_e=+1, h_p=+1, h_\gamma=-1$   
electron L, proton h=-1

Leading initial-ensemble/final-state components (N=2, retained=99.8%)



# L auxiliary lepton characteristic max $C_{p\gamma}$ configuration

3D momenta

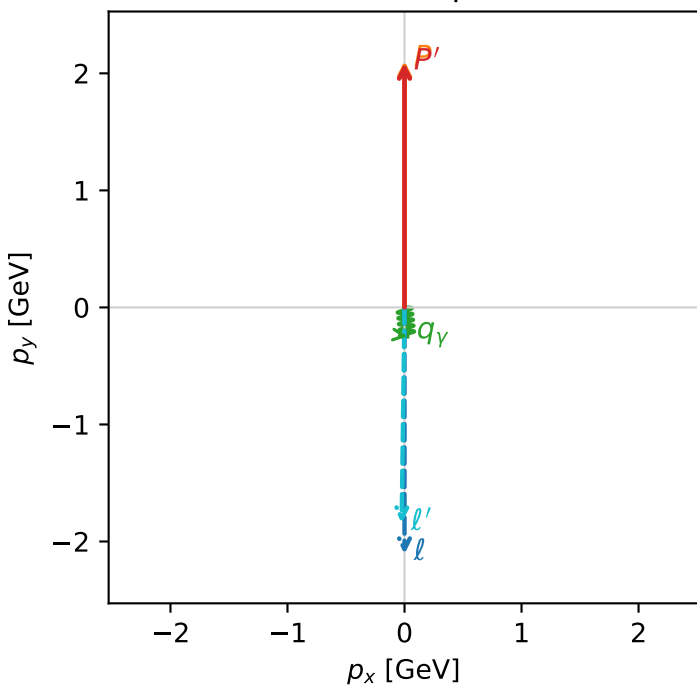


c\_p\_gamma\_low\_egamma\_l\_lepton\_region\_3  $C_{p\gamma}=0.0119042$   
 region: low\_Egamma\_L\_lepton\_region\_3 final-pair delta\_xy=3.04342 rad  
 outgoing-state purity: 0.500722  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.10694$ ,  $|\vec{P}'|=2.09128$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=4.81056$   
 $Q^2=0.0149831$ ,  $x_B=0.00680558$ ,  $t=-4.08254e-05$ ,  $W^2=3.06645$ ,  $y=0.11212$   
 $\theta(\ell', \gamma)=6.38697$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.3322$   $p_x= -0.0000$   $p_y= -2.1069$   $p_z= -0.0000$   
 $P$   $E= 2.3063$   $p_x= 0.0000$   $p_y= 2.1069$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0965$   $p_x= -0.0245$   $p_y= -1.8425$   $p_z= 0.0000$   
 $P'$   $E= 2.2920$   $p_x= 0.0000$   $p_y= 2.0913$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.0245$   $p_y= -0.2488$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=99.9%)

